722. CLINICAL ALLOGENEIC TRANSPLANTATION: ACUTE AND CHRONIC GVHD, IMMUNE RECONSTITUTION | DECEMBER 06, 2014

Probiotic Enteric Regimen for Easing the Complications of Transplant

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Abstract

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Background

Since the recognition of the important role the intestinal microbiome is playing in our well being, research of its complex nature is gaining momentum. Early recognition for the potential of dysbiosis may help in taking measures to ward off the microbial imbalance, and thus prevent untoward clinical disorders. Preliminary data in animals has indicated that the administration of probiotics may modify graft versus host disease (GVHD) following allogeneic hematopoietic stem cell transplant. In humans, recent research has suggested that probiotics may favorably impact gut related immunity. We performed a pilot study to determine the safety of administering the probiotic *lactobacillus GG* following hematopoietic engraftment after allogeneic hematopoietic stem cell transplant (HSCT). There were no untoward events associated with administration of the probiotic. This retrospective analysis was designed to analyze the effects of the probiotic on the incidence of intestinal GVHD among patients participating in the study.

Methods

We evaluated eighty patients who underwent an allogeneic HSCT between 1/3/2006 and 9/13/2013 through our bone marrow transplant registry. Patients were classified into two groups. Thirty patients were

supplemented with the probiotic *lactobacillus GG* as part of the pilot study, and fifty patients not receiving probiotic therapy were chosen as historic controls. The primary end point of this retrospective analysis was the development of acute GVHD (aGVHD) or chronic GVHD (cGVHD). Patients were evaluated at three, six and twelve months following HSCT.

Results

Among the thirty patients who were treated with probiotics, at the three-month interval, 14 (47%) developed stage I aGVHD, 6 (20%) developed stage II aGVHD, and 10 (33%) were free of disease. At six months, 12 patients (40%) had limited cGVHD, 5 (17%) had extensive cGVHD, and 13 (43%) were without disease. At 12 months, 12 patients (40%) had limited cGVHD, 7 (23%) had extensive cGVHD, 7 (23%) were free of disease, and 4 (13%) were unverifiable.

For the fifty patients who did not receive probiotics, at the three-month interval, 21 (42%) developed stage I aGVHD, 5 (10%) had stage II, and 6 (12%) had stage III or IV aGVHD. Seventeen (34%) patients were free of disease, and 1 (2%) was unknown. At six months, 17 patients (34%) had limited cGVHD, 10 (20%) had extensive cGVHD, 19 (38%) were without disease, and four patients (8%) were unknown or expired. At the 12-month period, 10 patients (20%) had limited cGVHD, 21 (42%) had extensive cGVHD, 15 (30%) had no evidence of disease, and 4 (8%) were unknown or expired.

Conclusion

Our findings suggest that supplementing therapy with probiotics may modulate risk of GVHD with the potential for lower rates of stage III- IV aGVHD. However, this retrospective analysis does not prove that it reduces the overall incidence of GVHD in this patient population. This study demonstrates the safety and feasibility of administering probiotics in post-allogeneic stem cell transplant patients. A randomized study is now ongoing to determine the efficacy of probiotics with respect to aGVHD and cGVHD.

Disclosures

Off Label Use: Probiotics for the prophylaxis of GVHD.

Author notes

*Asterisk with author names denotes non-ASH members.